

Practice — Matter Is Made of Particles

Section A is interactive — tap an option or type and press Check. Sections B and C are written practice.

Objective (1 mark each)

Q1 • MCQ**[1]**

The smallest basic units that make up a substance are its —

- (a) interparticle spaces (b) constituent particles (c) mixtures (d) bubbles

Q2 • MCQ**[1]**

When sugar dissolves in water, it —

- (a) disappears completely (b) breaks into particles that fit between water particles
(c) becomes a new substance (d) floats on top

Q3 • MCQ**[1]**

Grinding chalk into powder is a —

- (a) chemical change (b) physical change (c) new substance (d) reaction

Q4 • MCQ**[1]**

The ancient Indian philosopher who proposed indivisible particles (Parmanu) was —

- (a) Aryabhata (b) Acharya Kanad (c) Charaka (d) Sushruta

Q5 • TRUE / FALSE**[1]**

Constituent particles of matter can be seen with an ordinary microscope.

- (a) True (b) False

Q6 • FILL IN THE BLANK**[1]**

The gaps between the particles of matter are called _____ spaces.

Q7 · FILL IN THE BLANK

[1]

We can sense dissolved sugar in water by its _____.

Short answer (2 marks each)

Q8

[2]

What are constituent particles? Give an example.

Q9

[2]

Explain, using particles, why sugar seems to disappear in water but the water tastes sweet.

Q10

[2]

Why is dissolving salt in water considered a physical change?

Long / application (3 marks each)

Q11

[3]

A student grinds chalk to a fine powder and another dissolves sugar in water. Explain how both activities support the idea that matter is made of tiny particles.

Q12

[3]

When sugar is added to a full glass of water it dissolves without the water overflowing much. Explain this observation.

[Check yourself](#) → Answer key